
SOFTWARE REQUIREMENT

Establishing the business requirements (chapter 05)

Objectives

- Understanding the business requirements and how to define them.
- Student should enhance the structure of vision and scope document, a deliverable that contains the project's business requirements
- Student could write the vision and scope about project through using vision statement and scope representation techniques.
- Understanding the vision and scope on agile project

Contents

- Business requirements
 - Identifying desired business benefits
 - Product vision and project scope
 - Conflicting business requirements
- Vision and scope document
- Scope representation techniques
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 - Ecosystem map
 - Feature tree
 - Event list
- Keeping the scope in focus
- Vision and scope on agile project

Business requirements

- A set of information that describes a need that leads to one or more projects to deliver a solution and the desired ultimate business outcomes.
- Making up the business requirements.
 - Business opportunities,
 - Business objectives,
 - Success metrics,
 - Vision statement

Business requirements

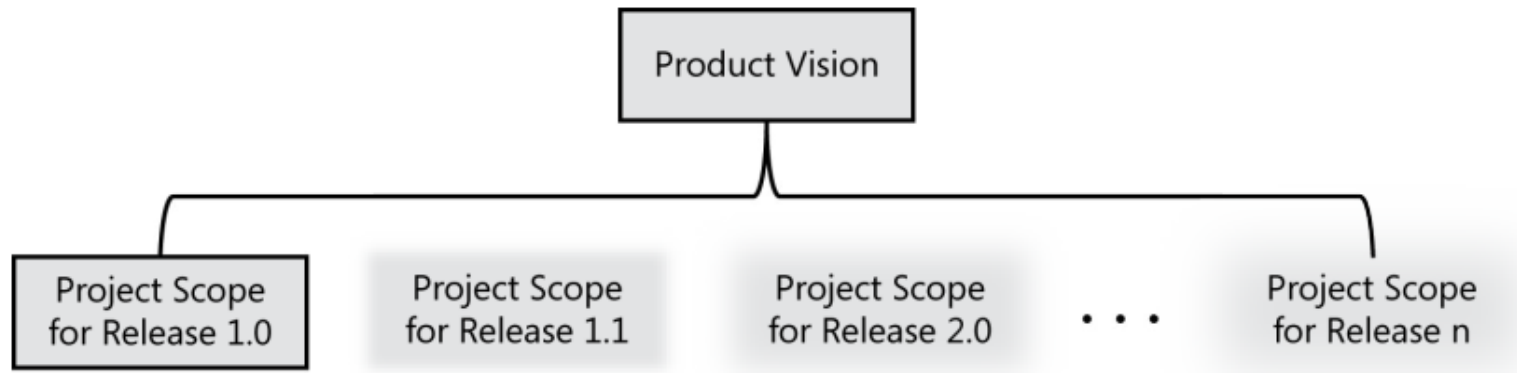
- Business requirements issues must be resolved before the functional & nonfunctional requirements can be fully specified
- A statement of the project's scope and limitations helps greatly with discussions of proposed features and target releases
- Provide a reference for making decisions about proposed requirement changes and enhancements
- Displaying the business objectives, vision, and scope highlights in every requirements elicitation session so the team can quickly judge whether a proposed requirement is in or out of scope.

Business requirements

- Identifying desired business benefits
 - The business requirements set the context for, and enable the measurement of, the benefits the business hopes to achieve from undertaking a project.
 - Business requirements might come from funding sponsors, corporate executives, marketing managers, or product visionaries.
 - The business benefit has to represent a true value for the project's sponsors and to the product's customers.

Business requirements

- Product vision and project scope
 - The **product vision** succinctly describes the product/solution that will achieve the business objectives. The vision describes what the product is about and what it ultimately could become.
 - The **project scope** identifies what portion of the ultimate product vision the current project or development iteration will address.



Business requirements

- Conflicting business requirements
 - Business requirements collected from multiple sources might conflict.
 - The project's decision makers must resolve the conflicts before the analyst can detail the requirements
 - Uncontrolled scope creep, in which stakeholders overstuff the new system in an attempt to satisfy every interest, can cause the project to topple under its own weight.
 - A BA can help by surfacing potential areas of conflict and differing assumptions, flagging conflicting business objectives, noting when requested features don't achieve those objectives, and facilitating conflict resolution
 - Long-duration projects often experience a change in decision makers partway through

Vision and scope document

- Collects the business requirements into a single deliverable that sets the stage for the subsequent development work.

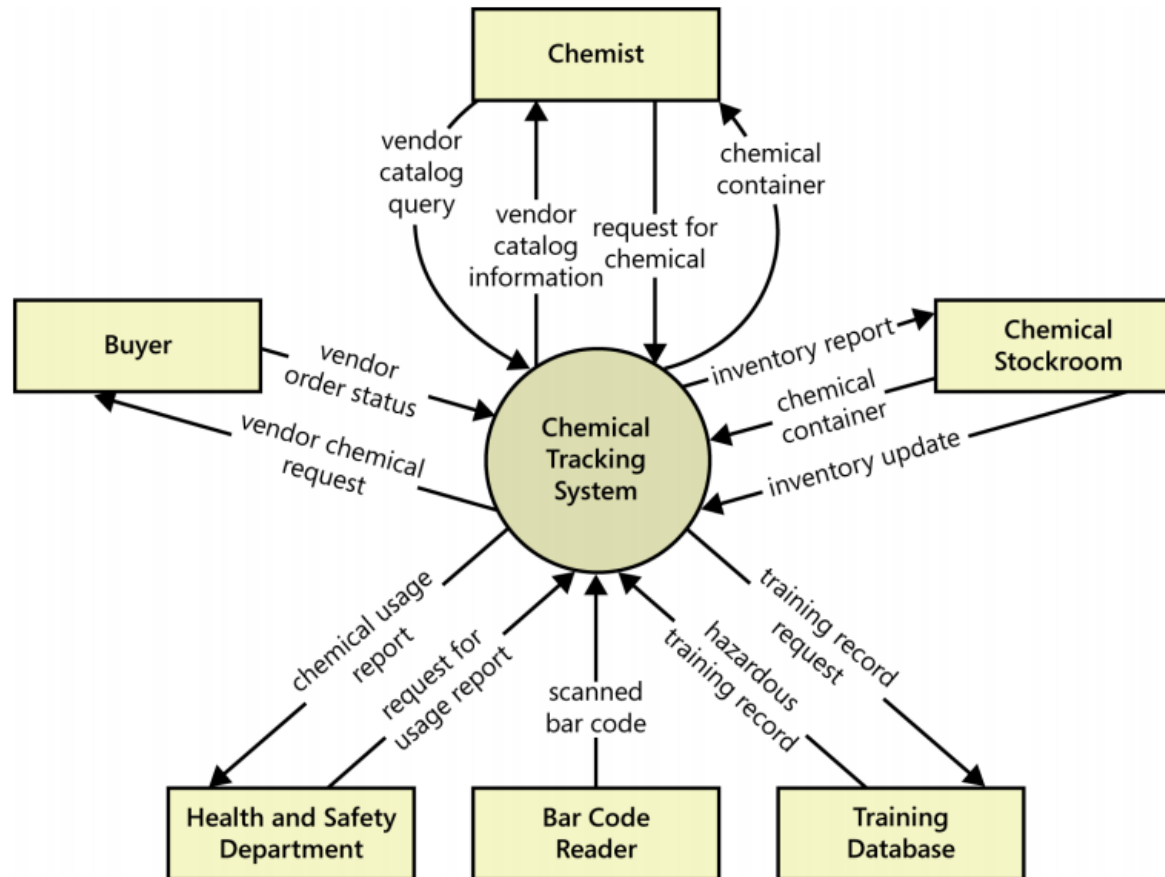
- A business case document
- Project charter

- The owner of the vision and scope document is the project's executive sponsor, funding authority, or someone in a similar role. A business analyst can work with this individual to articulate the business requirements and write the vision and scope document.

1. Business requirements
 - 1.1 Background
 - 1.2 Business opportunity
 - 1.3 Business objectives
 - 1.4 Success metrics
 - 1.5 Vision statement
 - 1.6 Business risks
 - 1.7 Business assumptions and dependencies
2. Scope and limitations
 - 2.1 Major features
 - 2.2 Scope of initial release
 - 2.3 Scope of subsequent releases
 - 2.4 Limitations and exclusions
3. Business context
 - 3.1 Stakeholder profiles
 - 3.2 Project priorities
 - 3.3 Deployment considerations

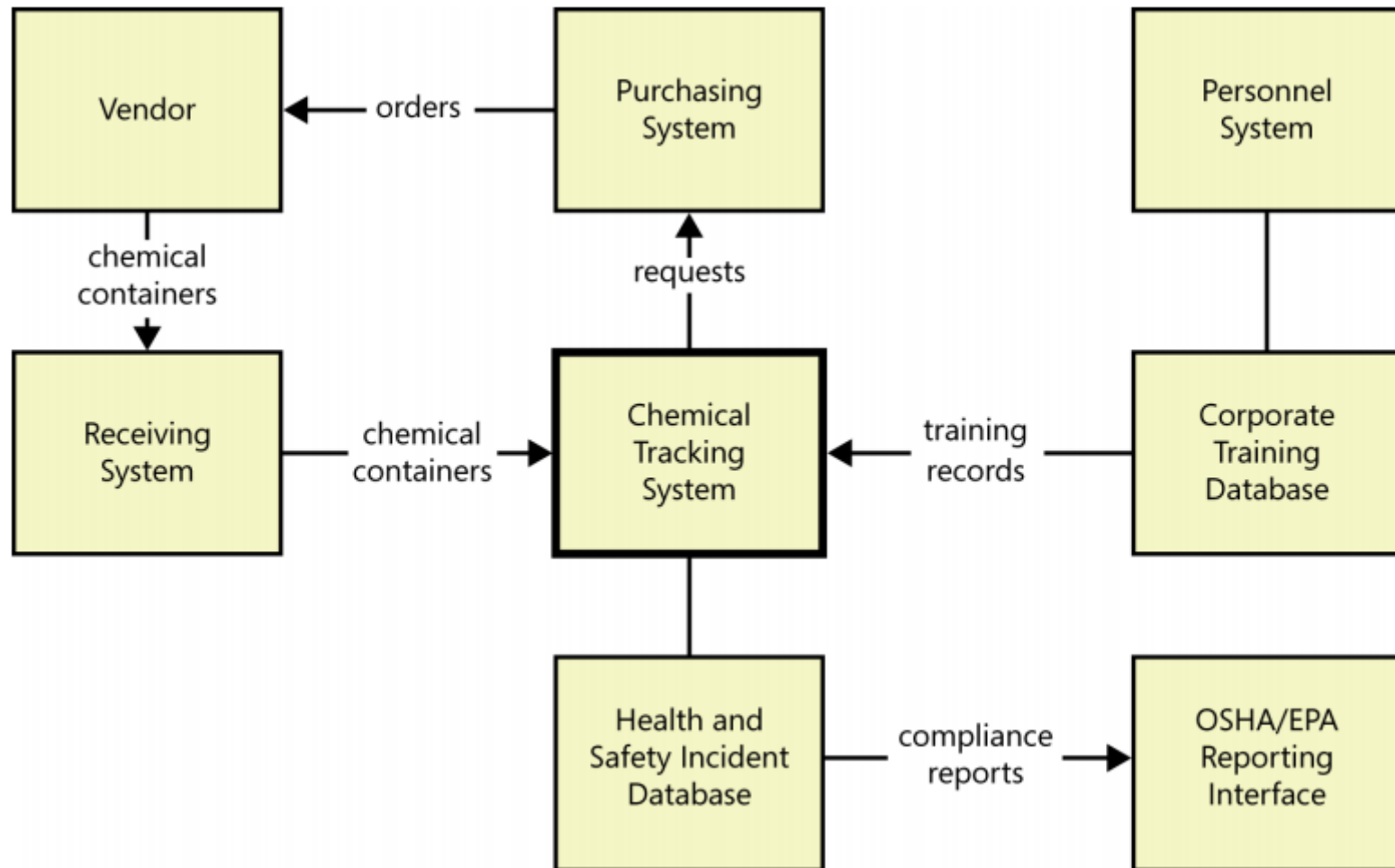
Scope representation techniques

Context diagram



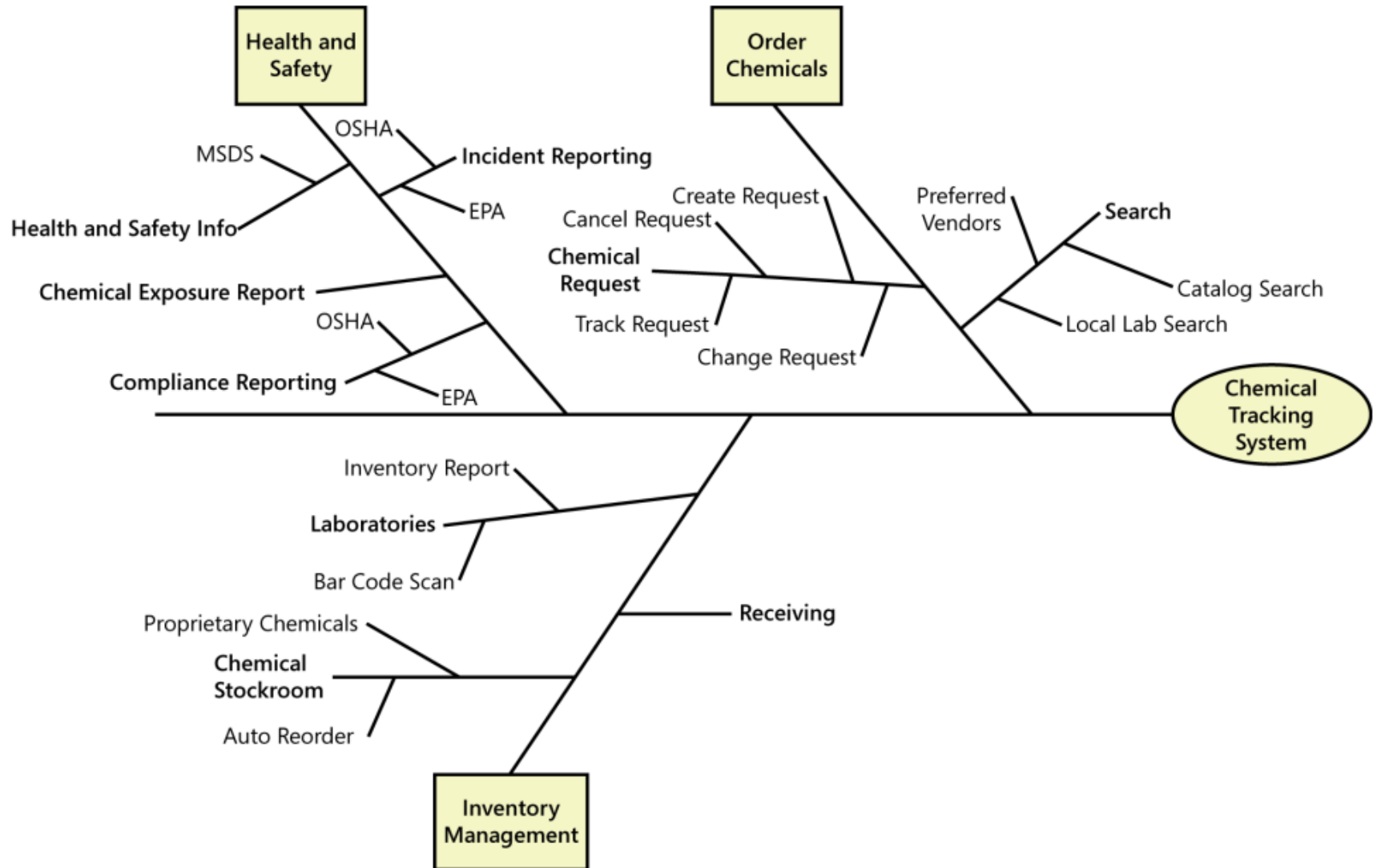
Scope representation techniques

Ecosystem map



Scope representation techniques

Feature tree



Scope representation techniques

Event list

- An event list identifies external events that could trigger behavior in the system.
- It depicts the scope boundary for the system by naming possible business events triggered by users, time-triggered (temporal) events, or signal events received from external components, such as hardware devices.

External Events for Chemical Tracking System

- Chemist places a chemical request.
- Chemical container bar code is scanned.
- Time to generate OSHA compliance report arrives.
- Vendor issues new chemical catalog.
- New proprietary chemical is accessioned into system.
- Vendor indicates chemical is backordered.
- Chemist asks to generate his chemical exposure report.
- Updated material safety datasheet is received from EPA.
- New vendor is added to preferred vendor list.
- Chemical container is received from vendor.

FIGURE 5-9 Partial event list for the Chemical Tracking System.

Keeping the scope in focus

- The business requirements and an understanding of how customers will use the product provide valuable tools for dealing with scope change
 - Change is within the scope -> priority
 - Change is out of scope & for future release or by another project only
 - Change is out of scope, but it's such a good idea that the scope should be broadened to accommodate it, with corresponding changes in budget, schedule, and/or staff
- The information in the vision and scope document lets you assess whether proposed requirements are appropriate for inclusion in the project

Keeping the scope in focus

- Using business objectives to make scoping decisions
 - The business objectives are the most important factor to consider when making scope decisions.
 - Determine which proposed features or user requirements add the most value with respect to the business objectives; schedule those for the early releases.
 - When a stakeholder wants to add functionality, consider how the suggested changes will contribute to achieving the business objectives.
 - If possible, quantify the contribution the feature makes towards the business objectives, so that people can make scoping decisions on the basis of facts rather than emotions

Keeping the scope in focus

- Assessing the impact of scope changes
 - A common consequence of scope change is that completed activities must be reworked in response to the changes.

Vision and scope on agile projects

- The scope of each iteration consists of user stories selected from a dynamic product backlog (Priority, Team capacity)
- Instead of trying to fight scope creep, the team prioritizes new requirements against existing items in the backlog and allocates them to future iterations
- Overall **project duration** still depends on the total amount of functionality to be implemented, but the scope of each iteration is controlled to ensure timely completion
- The team can define a high-level roadmap of iterations at the beginning of the project, but the user story allocation for an iteration will be performed at the beginning of each iteration (Referencing the business requirements)